

You've harvested a ton of delicious crops – now it's time to clean them up! Did you know that, on average, a typical small-scale mixed vegetable farm spends 40% or more of their labour on harvesting, washing and handling their produce? The wash/pack station is the setting where this work happens, yet often enough, growers don't invest much time or money considering it.

Here is a list of 8 design principles that will help you increase the efficiency, workflow and food safety of your post-harvest handling.

COVER YOUR WALLS AND CEILINGS

The first thing to consider when setting up your wash/pack station is to make sure that your walls and ceiling are smooth, cleanable, and impermeable to water. This is your first line of defence against mold and bacteria. There are several commercial options out there but whatever you decide, it should be:

- **Moisture Proof**
- **Maintenance Free**
- **Rot/Rust/Corrosion Proof**

Commonly used surfaces are vinyl or PVC-based, which can be installed directly over most solid surfaces, such as wood, concrete, drywall and many other solid surfaces.. These panels allow a certain portion of the light to enter into the grow room through reflection, providing a bright room as you wash and pack your crops.

Many growers use sheet metal roofing panels, which also can work, but screws popping out make these panels harder to wipe clean. When it comes down to it, most washable and impermeable wall fabrics can also provide an effective barrier. Again, you are looking for a smooth, hygienic, abuse-resistant finish that is easy to clean and maintain.

INVEST IN CEMENT FLOORING

Covering your floor with cement is probably the most expensive component of your washing/packing area, but is also the investment that can provide the biggest return. Having a smooth, slick surface to work on allows you to properly sweep your space, keeping your floor neat and clean. The wash water can also be easily handled by using a

squeegee. Cementing your floor will allow you to use different transport tools to move your product around (see Principle 3).

Using cement brings up two important factors to consider before starting the work:

- **Drainage:** If left to sit, your wash water can harbor pathogens that can compromise the food safety in your wash/pack station. Proper drainage is essential. Your cement floor should slope towards a drain. Depending on the size of your washing area, this drain could be in the middle or on the end wall, below your gunning station. Because your drain will be collecting lots of soil particles as you clean your veggies, it's critical to design it with a sedimentation tank that you can clean, if filled. A proper dispersal method is also something to think about.
- **Non-Slippery:** Wet cement floors can be very hazardous if they become slippery. To counter that, make sure to keep a rough texture.

MINIMIZE LIFTING BY GOING MOBILE

Relying on wheels to transport your produce across cement flooring will save you time and energy. Your goal should always be to minimize lifting and hand carrying and always work by stacking containers that move more veggies at once.

- **Hand-trucks or dollies** can allow you to move and stack containers, without requiring a smooth, leveled floor to operate. However, food safety rules require that harvested produce, once washed and packed, should never touch the floor. The way around this is to rely on using mini-pallets that your hand truck will be able to pick up.
- **Skate wheel tracks or scooters** are very simple, yet effective. They allow you to stack your produce boxes and wheel them freely and simply. Their low cost makes it worth it to purchase many at once to have available in or near your wash/pack station.
- **U-boat** are great at wheeling lots of harvest containers at once. We use them to load up the harvest that is coming in from the field and transporting it to the wash station. We also use them to quickly deliver produce into the cooler to pre-chill certain veggies when need be.
- **Pallet jacks** are great to move big loads around, especially material that comes in pallet deliveries (soil mix, fertilizers, etc.). Having the option to wheel those inside your packing area is a plus. All access points, such as doors and corridors should be designed for the potential use of a pallet jack, with room to steer in and out.
- **Make it easy to reconfigure** your wash/pack area by ensuring all structures are on wheels (tables, packing shelves, root washer, etc.). The goal is to create different working spaces from the same space, allowing you to use the setting more efficiently. This option also diminishes the risk of making mistakes when designing the workflow of the wash/pack area.

AMP UP YOUR WATER DELIVERY

One of the (if not the key) to efficiency in your wash/pack station is having reliable access to potable (safe to drink) water for cleaning your veggies. The quality of the water is a critical issue for any food safety standards and it's your responsibility to perform periodic water sampling and microbial testing. The flow rate is also key, since it determines your overall capacity to fill your tubs and feed your gunning stations. Having the adequate plumbing for the flow needed is the solution.

Once your quality and input are sufficient, then focus on ways to quickly wash out soil when you are gunning your bunches or filling tubs. In both cases, maintaining high water flow is key.

- **Gunning stations** should have high pressure, and the way to achieve this is by concentrating the flow in your piping. The main line coming from your source (pump, well, city, etc.) should be as big as possible. From this main line, connect a smaller line that attaches to your washing gun. For example, if your main supply line is 1" then your water gun line should be 3/4 inch. This will concentrate the flow and bring more pressure to your gun. The choice of gun can also make a difference, so choose one that concentrates the flow.
- **Tubs** need less pressure and more volume to quickly fill up. In the example above, you bring your 1" line all the water to your tub. The piping should allow you to fill more than one tub at a time or simultaneously with your gunning station. By installing inexpensive floating valves, your tubs can be filled while you work on something else.
- **A bubbler** can improve the efficiency of your water usage by eliminating all double washing of your greens.
- **Consider water delivery from above.** If using hoses, installing hoses towards the ceiling keeps them cleaner and up off the ground, which also minimizes tripping hazards and moisture on the floor.

SANITIZE

In order to prevent food-borne illness, sanitization is critical for any post-harvest handling system and it starts in the wash/pack area. The water you use to wash vegetables must be clean and potable.

- **Wash up!** Anyone working in the post-harvest area should thoroughly wash their hands prior to handling or washing any produce. Installing a simple hand-washing station in your wash/pack area isn't mandatory but highly recommended. If that is not possible,
- **Sanitize** all surface areas where clean veggies touch a surface area, and particularly, your tables. This is especially true when bagging greens. The process starts with

cleaning any soil and organic matter from the produce and making sure that your tables and handling areas are properly sanitized. You can do this with bleach solutions, but you are better off with using a peroxyacetic acid-based sanitizer. When handling any of these products make sure to protect your eyes.

- **The choice of work tables** is also important. Smooth surfaces allow easy cleaning. Rough surfaces harbor dirt and microorganisms. Anything wood should be avoided, just like anything that can rust. The best option is stainless steel and cheap tables can be found on sites that sell used restaurant equipment.
- **Rodents, including all animals, cats and dogs cannot access the wash/pack area.** Plastic bags and box liners should be kept in sealed storage containers. Rodent bait stations should be set throughout the building and regularly monitored for activity.

ERGONOMICS

Picking up heavy things and doing repetitive work in postures that aren't ergonomically friendly can lead to chronic fatigue or injury. Therefore, it's important that you design your wash/pack area with ergonomics in mind.

- **The height of working stations should be around 30".** Have an option for smaller or taller workers if needs be.
- **Regulate your harvest and storage containers** by picking formats that won't be too heavy when lifting. Have one bigger size for your greens and another average size for your bunches. If you're dealing with a large harvest of different roots (potatoes, rutabaga, etc.) this is especially important.
- **A loading ramp/dock** can help you avoid manipulating each produce box individually when delivering. The process of loading your produce should be very safe as it presents a safety hazard.

MAKE IT BRIGHT, FUN AND NEAT

Your wash/pack station is where you spend many hours in a given week, so make it a neat and joyful space!

- **Windows** make life better, but keep in mind that your wash pack area should be kept cool at all times. Considerations include: placing windows on the north wall where they won't let direct sunlight in; using screened windows that can open without letting bugs and birds enter the wash station
- **Garage or large doors** are fun because they make you feel like you're outside, but if you're using them to let light in, make sure to keep birds out. On hot days, the coolness of the room can also become compromised if the doors are left open for too long.

- **Use lots of lights, but keep them safe.** Fluorescent lights (shop lights) provide the best option and should be installed right above every workstation (including gunning stations, tubs, bubbler etc.). To limit impacts in case of breakage, each bulb should be enclosed with a plastic shield.
- **Music can help to provide a positive environment** when you/your staff are washing veggies for hours on end. I recommend using a sound system over using individual earbuds. Karaoke challenges can provide good team spirit.
- **Cleanliness is key.** Keep it clean to keep it clean. Each wash session should end with clean-up. Harvest containers should be washed and stacked to dry out. The gunning station and any tubs used should be emptied of organic matter and cleaned. The soil or mud on the floor should be swept and washed.
- **Aprons, rubber boots and gloves** should have a space to dry between sessions. Having extra pairs of each is also important to have available for when others get wet.
- **Establish a ‘command centre’** where harvest tallying, order taking and invoice printing can occur until it’s time to wash or pack veggies. Make this space waterproof.

THINK IN WORKFLOW

Considering your wash/pack area from a bird’s eye view and following the trail of where the veggies travel from one area to another can help to identify all the extra-steps and wasted time in your process. This workflow can be understood by drawing a simple diagram of your wash/pack area. You can also follow someone from your crew during the whole sequence to better understand where the bottlenecks may be. Some key considerations:

- **Cross contamination.** Your workflow should be arranged so that the product moves to a cleaner area during each step of processing. In this way, one directional flow (dirty-clean- delivery) is optimal over a U-shape.
- **Access doors** should be big enough to allow wheel carts to easily enter and leave rooms 4’ wide doors are plenty for easy maneuverability. If installing garage doors, include a main door besides it.
- **Storage supplies** such as bags, rubber bands, scales and boxes should be kept close to where they are needed. All items or equipment not used in the wash/pack area should be removed from the area. This area should be vermin-free.
- **Order area.** Packing produce in an unsystematic manner wastes time and effort. Use multilevel shelves with stackable functions. Every ergonomic detail counts.
- **Eliminate extra handling** by sorting the veggies for each sales channel when washing them.

- **The bagging area** is where greens or other veggies are handled. This area needs to be thoroughly sanitized and workers need to wash their hands prior to starting any work. The set up should be adjacent to the spinning machines to minimize steps. Plan to create a shared workstation that can have more than 1-person bagging. ‘Dirty’ veggies should not come into contact with this area.
- **The cold room** is where the veggies are refrigerated and cooled down. Different veggies store better at different temperatures, so having more than one cold room is essential for maintaining optimal quality of the veggies and bigger is always better. It should be as big as possible to allow comfortable circulation and proper air flow between produce containers.

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